PROJECT OUTLINE

Introduction

South Africa bears a dual epidemic of HIV and tuberculosis (TB), with 7.8 million people living with HIV and approximately 360,000 new TB cases annually, 58% of which are HIV-positive (Thembisa 4.8, 2025). TB causes around 50,000 deaths yearly, 46% among HIV-positive individuals, underscoring the need for targeted interventions (WHO Global TB Report, 2024). Suboptimal TB treatment outcomes in HIV-positive patients, driven by factors like drug resistance and poor antiretroviral therapy (ART) adherence, challenge the National Strategic Plan for HIV, TB, and STIs (2023–2028) goal of 95% TB treatment success. This study proposes a predictive biostatistical model to identify factors influencing TB treatment success in HIV-positive patients, using national health data to optimize interventions. It aligns with NRF priorities of health and well-being, technology and innovation, addressing inequality, and human capacity development by tackling a major public health issue, employing advanced modeling, and focusing on a vulnerable population.

Problem Statement

Despite integrated HIV/TB services, TB treatment success rates in HIV-positive patients remain below targets (87% nationally, NICD, 2023), with gaps in predicting outcomes at a national level. Current studies often use descriptive analyses or outdated data (e.g., 2018 General Household Survey), limiting their predictive power. Few leverage clinic-level data like DHIS2 or employ machine-learning techniques to model complex HIV/TB interactions. This project addresses this gap by developing a predictive model for TB treatment success, using recent data (2019–2023) and innovative methods to inform policy and improve outcomes.

Aims and Objectives

Aim: To develop a biostatistical model predicting TB treatment success in HIV-positive patients in South Africa to enhance intervention strategies.

Objectives:

- Identify key predictors of TB treatment success (e.g., TB type, ART adherence, CD4 count) using national health data.

- Build and validate a predictive model using multilevel logistic regression and random forest techniques.

- Assess the model's policy implications for the National TB Programme and resource allocation.

Methodology

Study Design: A retrospective cohort study using national TB/HIV data from 2019–2023.

Data Sources:

Primary: DHIS2 (accessed via MSc institution, e.g., University of Pretoria, post-2026 enrollment) for patient-level data on TB treatment outcome (binary: cured/completed vs. failed/died/lost), TB type (drug-sensitive/MDR-TB), TB treatment duration (months), previous TB history (yes/no), sputum smear status (positive/negative), ART adherence (%), CD4 count (<200 vs. ≥200 cells/µL), age, gender, and clinic type (urban/rural).

Alternatives: If DHIS2 access is restricted, patient-level data will be sought from NICD TB surveillance (e.g., cohort data) or SAMRC cohorts (e.g., 2018 TB Prevalence Survey). Open cohort datasets (e.g., CAPRISA, via zenodo.org) are also explored, guided by a former professor (emailed June 3, 2025).

Contingency: If only aggregate data is available, an ecological study will use WHO Health Indicators data.humdata.org/dataset/who-data-for-zaf and Thembisa 4.8 thembisa.org for provincial TB success rates, HIV prevalence (58%), ART coverage (74%), and health workforce metrics.
Variables:

Outcome: TB treatment success (binary).

Predictors: TB type, treatment duration, previous TB history, sputum smear status, ART adherence, CD4 count, age, gender, clinic type, province, health facility resources (WHO).
Analysis:

Multilevel Logistic Regression: To account for clinic-level clustering (e.g., urban vs. rural facilities).

Random Forests: To capture non-linear relationships and rank predictor importance.

Software: R (tidyverse, caret packages), practiced via TidyTuesday TB dataset (GitHub, June 11, 2025).

Validation: Sensitivity/specificity, cross-validation to prevent overfitting.

Missing Data: Multiple imputation for incomplete records (e.g., ART adherence).
Ethics: Approval will be sought from the University of Pretoria Ethics Committee, ensuring data privacy.

Expected Impact
The predictive model will identify high-risk HIV-positive TB patients, informing targeted interventions under the National Strategic Plan (2023–2028). By optimizing ART adherence and MDR-TB management, it supports the 95% TB success goal, reducing mortality and healthcare costs for the National Health Insurance. Findings will be shared with SAMRC and the Department of Health to enhance TB/HIV policy. The project builds biostatistical capacity, equipping the applicant to address South Africa's health challenges, aligning with NRF's human capacity development goal.

Addressing Inequality
Focusing on HIV-positive TB patients, a vulnerable group disproportionately affected in provinces like KwaZulu-Natal (27% HIV prevalence, HSRC, 2023), this study tackles health disparities. A national model ensures equitable resource allocation, with potential for future province-specific analyses (e.g., KZN), addressing NRF's inequality priority.

Innovation
Using random forests alongside logistic regression introduces machine-learning innovation, rare in South African TB/HIV studies. Leveraging recent data (2019–2023) and open-source R tools enhances reproducibility and scalability, aligning with NRF’s technology and innovation focus.

Conclusion
This project addresses a critical gap in predicting TB treatment outcomes in HIV-positive patients, using innovative biostatistical models and recent national data. By aligning with NRF priorities—health, innovation, equity, and capacity building—it offers significant policy impact and personal development. Contingency plans ensure feasibility despite data access challenges, making it a competitive candidate for NRF funding.